



Target Users: District Magistrates (DMs), Vigilance Officers, Members of Parliament (MPs)

The Problem: The Rs. 10 Lakh Crore Leak in Public Works

India spends over **Rs. 10 Lakh Crore (\$120B+)** annually on rural and urban infrastructure (PMGSY roads, Jal Jeevan Mission drinking water, MPLADS community facilities). However, physical verification before treasury disbursement is completely manual.

1. Recycled / Staged Evidence	2. Ghost Roads & Distance Violations	3. Schedule of Rates (SoR) Inflation
Contractors submit photos of older completed roads or adjacent projects to claim milestone payouts before actual execution.	Funds drawn for unexecuted works or projects executed kilometers away from the sanctioned alignment with no geo-verification.	Invoices billed at inflated rates above government ceilings (e.g. Bitumen billed at Rs. 88k/MT vs Rs. 65k/MT approved SoR).

The Solution: Nirman-Drishti Core AI Engine

Nirman-Drishti functions as an automated digital vigilance auditor that runs **3 instant multimodal checks in under 5 seconds** before any public money leaves the exchequer.

Check 1: Geospatial Audit	Check 2: Gemini 2.0 Vision	Check 3: Rate Reconciliation
• Haversine distance calculation from tender coordinates. • Extracts EXIF metadata directly from evidence image. • Flags geofence perimeter breaches.	• Gemini Multimodal Vision stage detection. • Identifies raw gravel (WBM) vs finished bituminous wear coat. • Flags structural defects & missing layers.	• Line-by-line comparison with State Schedule of Rates. • Quantity ceiling verification against sanction. • Flags phantom/unsanctioned items automatically.

Real-World Benchmark Results (Demonstrated Cases)

Nirman-Drishti was tested against realistic public works claims across multiple government schemes:

Scenario / Tender	Claimed Amount	AI Forensic Findings	Verdict
PMGSY Rural Road (Honnavar to Karki, 1.8km)	Rs. 48.0 Lakhs	• Photo taken 11.5 km outside tender geofence. • Visual evidence shows raw gravel (tar missing). • Bitumen billed at Rs. 88k/MT vs Rs. 65k/MT SoR. • Phantom item of Rs. 2.25L detected.	REJECTED Risk: 100/100 Saved: Rs. 48 Lakhs
Jal Jeevan Mission (Solar Pump, PHC Shirwal)	Rs. 14.5 Lakhs	• GPS coordinates matched within 28 meters. • Solar array, 5HP pump, and tank verified operational. • Line-item rates match approved SoR exactly.	APPROVED Risk: 0/100 Disbursed: Rs. 14.5L
MPLADS Community Center (Najafgarh Skill Center)	Rs. 32.5 Lakhs	• Frame stage verified on site. • Rate inflation detected on RCC M25 concrete (billed Rs. 8,200 vs Rs. 6,400 approved).	VIGILANCE HOLD Risk: 15/100 Withheld: Rs. 3.15L

System Architecture & Google Tech Stack

Multimodal AI Layer	Google Gemini 2.0 Flash / Pro API	Spatial vision inspection, layer identification, and automated discrepancy explanation.
Geospatial Layer	Google Maps / OpenStreetMap / Folium	GIS mapping of constituency tenders, geofencing, and distance offset verification.
Administrative Cockpit	Streamlit, Python, HTML/CSS Engine	Dual-theme (Light/Dark) operational dashboard for District Collectors and MPs.
DPI & Trust Layer	ReportLab, SHA-256 Cryptographic Engine	Official Government of India PDF inspection certificates with cryptographic tamper seals.

Strategic Value: Why Nirman-Drishti Wins

Direct Fit for Indian Governance	Zero-Competition Blue Ocean	Production-Grade Ready
Tailor-made for the authorities (District Magistrates, MPs, Vigilance Directors) evaluating the challenge.	Bypasses common 'pothole reporting chatbots' to solve the real trillion-rupee contracting and verification bottleneck.	Complete with 100% passing test assertions, official PDF certificate generation, and live dual-theme dashboard.

Repository: <https://github.com/hrlpavan/nirman-drishti> • **Track:** AI for Digital Public Infrastructure & Governance